

Deep Residual Injection for Full-Spectrum Forensic Signal Perception in Multimodal Large Language Models

Kaiqing Lin^{*1,2} Zhiyuan Yan^{*3} Ruoxin Chen^{*2} Ke-Yue Zhang^{†2} Yue Zhou¹ Caiyong Piao⁴ Bin Li^{✉1}
Taiping Yao² Bo Wang² Youchang Xiao² Shouhong Ding²

Abstract

Multimodal large language models (MLLMs) have been increasingly adopted in forensics for their robust semantic understanding. As AI-generated images become realistic, semantic-level inconsistencies alone are often insufficient for reliable detection. This motivates a critical question: *whether MLLMs can achieve full-spectrum forensic signal perception, i.e., capturing low-level generator artifacts without sacrificing pre-trained semantic knowledge*. We further perform a layer-wise analysis of forensic signal perception in MLLMs, showing that semantic information is primarily formed in the early-to-middle layers, whereas direct fine-tuning for artifact learning disrupts these semantic representations. Based on this insight, we propose Deep Visual Residual MLLM (Deep-VRM) to *preserve early semantic processing while injecting artifact-specific visual signals as a residual path into an intermediate layer*, where they are fused with semantic token representations and propagated through subsequent trainable layers. This enables later layers to jointly model semantic reasoning and signal-level forensic cues, and surprisingly, the model learns to adaptively leverage different levels of forensic signals depending on the input, achieving robust and generalizable detection performance. Extensive experiments show that our method achieves state-of-the-art across most benchmarks. The code and data are available at <https://github.com/KQL11/Deep-VRM>.

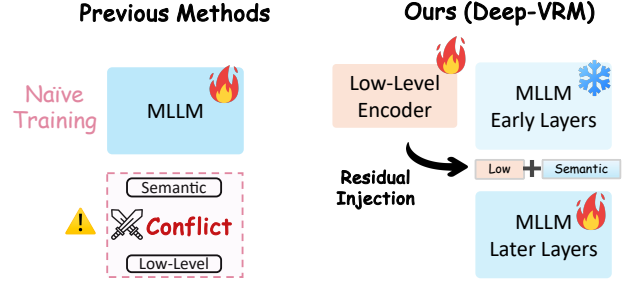


Figure 1. Comparison between naively training MLLMs for detection and ours. While previous methods suffer from a representation conflict between semantic and low-level features, our approach decouples these two spaces via residual injection. It preserves established semantic knowledge while introducing artifact sensitivity to resolve the learning dilemma.

1. Introduction

The growth of AI-generated content has precipitated a crisis in digital trust, raising urgent concerns regarding image authenticity. Numerous studies (Kang et al., 2025; Gao et al., 2025; Wen et al., 2025; Lin et al., 2025c; He et al., 2025) have proposed leveraging Multimodal Large Language Models (MLLMs) to detect AI-generated images and provide human-aware textual explanations. While MLLMs appear to be ideal candidates for detection due to their reasoning capabilities and interpretability, they face significant challenges in practice.

Generic MLLMs frequently struggle with the detection task itself, often underperforming compared to specialized forensic models (Jia et al., 2024). Consequently, current solutions often resort to external expert modules (i.e. specialized forensic models) to compensate (Chen et al., 2025b; Zhou et al., 2025b). Yet, this reliance reduces the MLLM to a mere proxy rather than an independent evaluator, preventing it from learning the intrinsic features of forgery. Moreover, such approaches fail to elucidate the underlying why there is bad performance of the MLLMs.

In this paper, our research reveals a critical trade-off in MLLM representation learning: **native models cannot learn generalizable generator traces without compro-**

^{*}Equal contribution [†]Project leader [✉]Corresponding author.

¹Guangdong Provincial Key Laboratory of Intelligent Information Processing, Shenzhen Key Laboratory of Media Security, and SZU-AFS Joint Innovation Center for AI Technology, Shenzhen University ²Tencent Youtu Lab ³Peking University ⁴Fudan University. Correspondence to: Bin Li <libin@szu.edu.cn>.

misgiving their core semantic capabilities, which leads to the above deficiency. Pretrained MLLMs are optimized for semantic features, such as image style and content and logic. Consequently, they inherently overlook low-level artifacts, specifically the subtle traces left by the generator, which are essential for forgery detection. We utilize two datasets, D_1 (focusing on semantic features) and D_2 (focusing on generator traces), to explore this mechanism. As shown in Figure 3, we find that standard LoRA tuning on the original MLLM architecture struggles to recover these low-level artifacts. Although updating more parameters (e.g., full fine-tuning) allows the model to learn these features, it damages its understanding of semantics (Table 2). Consequently, achieving the dual capability required for robust detection remains a significant challenge for current architectures.

To resolve this dilemma, we should understand how the model handles these features. Through a layer-wise analysis using linear probes, we observe that the separability of semantic features for discriminating real from fake images is primarily established and converges in the early-to-middle layers (see Figure 4). **This suggests that these layers are critical for extracting semantic features, while forcing these layers to learn low-level artifacts interferes with extracting semantic features in AIGI detection.** We observe that previous MLLM-based AIGI detection methods typically process low-level artifacts and semantic features in the same manner without distinction, resulting in suboptimal performance. Therefore, a strategy is to decouple the learning process: keeping the early layers frozen for semantics and injecting low-level artifacts in the later layers.

Guided by these insights and recognizing that the rich semantic priors of MLLMs provide a powerful foundation for detection, we propose the Deep Visual Residual MLLM (Deep-VRM), *focusing on enhancing the detection performance of MLLMs, distinct from the aspect of explainability.* As shown in Figure 1, our method employs a Residual Injection strategy to bypass the early layers where semantic features are encoded. We utilize a dual-branch visual architecture. The frozen encoder ($\mathcal{V}_o$) preserves the pretrained semantic perception already available in the MLLM, while the adapted branch ($\mathcal{V}_a$), equipped with LoRA adapters, learns artifact-specific visual cues. Unlike traditional methods that align features at the input level, Deep-VRM injects these artifact features directly into the deep layers of the LLM. Our method achieves state-of-the-art performance based on an MLLM only, avoiding the need for any external expert detectors. By learning low-level clues after the semantic features have converged, our model ensures robust performance in wild datasets.

- We systematically analyze the MLLMs in AIGI detection and reveal a critical *representation conflict*: forcing early-to-middle layers to learn low-level artifacts

compromises their inherent semantic capabilities. This finding establishes that effective detection requires decoupling artifact learning from semantic preservation.

- We propose **Deep Visual Residual MLLM (Deep-VRM)**, a novel architecture employing a *Residual Injection* strategy. Unlike previous methods, Deep-VRM injects artifact-specific features directly into the deep layers of the LLM, bypassing the semantic-dominant early layers to achieve optimal feature integration.
- Without external detectors, Deep-VRM achieves SOTA performance solely based on the MLLM. Extensive experiments demonstrate its superior generalization in-the-wild settings, effectively capturing both subtle generator traces and high-level semantic anomalies.

2. Related Work and Motivation

Traditional AIGI Detection Methods For AI-generated image (AIGI) detection, initial work CNNSpot (Wang et al., 2020) employed standard CNNs and conducted naive training. While effective in detecting samples from known generators, these classifiers often struggle to generalize to unseen ones. To address this, UnivFD (Ojha et al., 2023) utilizes CLIP as a backbone, leveraging the robust representations of pretrained vision-language models to enhance cross-generator generalization. Subsequent research (Liu et al., 2024; Tan et al., 2025; Zheng et al., 2024; Yan et al., 2025d; Tan et al., 2024b; Yan et al., 2025a;b; Shen et al., 2025; Lin et al., 2025a; Liu et al., 2026; 2025) has further refined this by exploring advanced architectures and preprocessing techniques. Some efforts (Tan et al., 2024a; Chu et al., 2025; Li et al., 2025a; Karageorgiou et al., 2025; Yan et al., 2026) focus on frequency-domain artifacts, identifying them as discriminative features for synthetic content. Moreover, recent works have explored pretraining-based strategies to further improve AIGI detection performance (Zhou et al., 2026; Li et al., 2025b), as well as calibration-based methods to enhance prediction reliability (Li et al., 2026; Guillaro et al., 2026). Despite these advancements, conventional detectors remain highly susceptible to image post-processing. In particular, their performance tends to degrade significantly when facing compressed images, raising concerns regarding their robustness in real-world scenarios.

MLLM-based AIGI Detection Methods Multimodal large language models (MLLMs) have demonstrated strong capabilities across downstream tasks, motivating their use for AIGI detection via instruction fine-tuning (IFT) (Kang et al., 2025; Gao et al., 2025; Wen et al., 2025; Lin et al., 2025c; He et al., 2025) and, more recently, reinforcement learning with self-exploration and verifiable rewards (Xu et al., 2025; 2026; Nguyen et al., 2025). By using image-text instructions, these methods guide MLLMs to identify

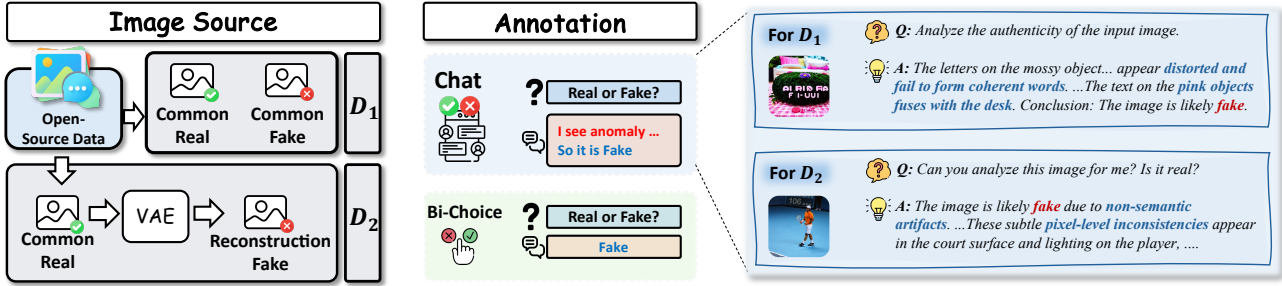


Figure 2. Illustration of the data construction pipeline. We first collect common real and fake images from various open-source datasets to form D_1 . Subsequently, we generate reconstructed fake images via VAEs and pair them with their original real counterparts to construct D_2 . To train the MLLM, we design two types of annotations—Chat and Bi-Choice—for all images in D_1 and D_2 . Notably, for the VAE-reconstructed images in D_2 , our annotations explicitly emphasize terms like “non-semantic artifacts” to guide the model in learning pixel-level discrepancies.

semantic inconsistencies or visual errors. However, MLLM-based detectors still struggle with generated images containing subtle low-level artifacts, such as generator traces. To mitigate this limitation, some frameworks integrate external expert detectors (Chen et al., 2025b; Zhou et al., 2025b; Peng et al., 2025), while others use MLLMs as agents to coordinate multiple experts (Zhu et al., 2026; Yu et al., 2026). Although these designs introduce external forensic knowledge, they may cause the MLLM to imitate expert predictions or rely on expert coordination, rather than developing autonomous feature-level visual analysis. Ultimately, prior research leaves two fundamental challenges unresolved: (1) how to enhance the perception of low-level forensic artifacts within the MLLM, and (2) how to balance the learning of low-level artifacts and semantic artifacts within the MLLM. In this work, we challenge the conventional end-to-end design of MLLM-based detection by considering the intrinsic functional heterogeneity of MLLM internal layers.

Exploration of Functional Stratification in MLLMs For large language models (LLMs), prior work (Skean et al., 2025) shows that early-layer representations are particularly sensitive to noise, whereas intermediate layers play a key role in processing complex semantic information by filtering redundant patterns and preserving discriminative features. In the context of MLLMs, recent studies (Jiang et al., 2025; Hartman et al., 2025) further identify intermediate layers as a critical locus for cross-modal interaction and visual reasoning. Together, these findings suggest a functional stratification inside LLMs and MLLMs: early-to-middle layers are crucial for forming stable semantic representations, while later layers are more responsible for high-level reasoning and information integration. However, how such layer-wise functional specialization affects forensic signal perception remains underexplored, particularly for low-level generator artifacts in AIGI detection. This motivates our analysis of how semantic cues and low-level artifacts are represented across MLLM layers.

Table 1. Source composition of our datasets D_1 and D_2 .

Dataset	Real Images			Fake Images			
	ImageNet	Ms-COCO	Unsplash	SD1.4 [†]	Echo 4o	Flux [*]	VAE
D_1	8500	0	5000	1000	250	3250	0
D_2	0	35000	0	0	0	0	35000

[†] sourced from Training Set of GenImage. ^{*} generated by Flux-dev.

3. Method

To address the model’s limited perception of low-level artifacts, we create a training dataset consisting of two distinct subsets. The first subset, D_1 , comprises diverse images aggregated from existing training sets (Russakovsky et al., 2015; Unsplash, 2025; Zhu et al., 2023; Ye et al., 2025) to capture common semantic cues for detection. Following the data construction protocol of DDA (Chen et al., 2025a), the second subset, D_2 , pairs original real images from MS-COCO (Lin et al., 2014) with their VAE-reconstructed counterparts generated using the Stable Diffusion 2.1 VAE. This approach eliminates semantic interference, forcing the model to focus exclusively on generative traces inherent in the reconstruction process. Both D_1 and D_2 are further formatted for instruction tuning. As illustrated in Figure 2, we construct two types of instruction annotations, applying both to datasets D_1 and D_2 . The first type follows a binary-choice format (bi-choice), consisting of a query such as “*Is this image real or fake? Please answer with only a single word: ‘real’ or ‘fake’.*” and a target response such as “*fake*”. For the second category, we utilized Gemini 2.5 Pro to generate detailed analytical annotations regarding image content and forgery traces. We follow the caption procedure proposed in the Forensic-Chat (Lin et al., 2025b). Notably, we also annotate the fake image in D_2 with some words like ‘Non-semantic artifacts’. By providing the model with ground-truth labels as priors, we ensured that the generated reasoning was both factually accurate. Detailed statistics are provided in Table 1. In total, we obtained 88,000 instruction-tuning samples.

Table 2. Impact of fine-tuning strategies on general multimodal benchmarks. Evaluated via VLMEvalKit (Duan et al., 2024) and Ms-Swift (Zhao et al., 2025), the results highlight a severe performance degradation in fully fine-tuned models, indicating catastrophic forgetting.

Model	BLINK	RealWorldVQA	MME
Backbone	0.5481	0.6758	1677
full finetuning LLM on D_2	0.0373	0.1137	506
Ours on D_2	0.5476	0.6721	1636

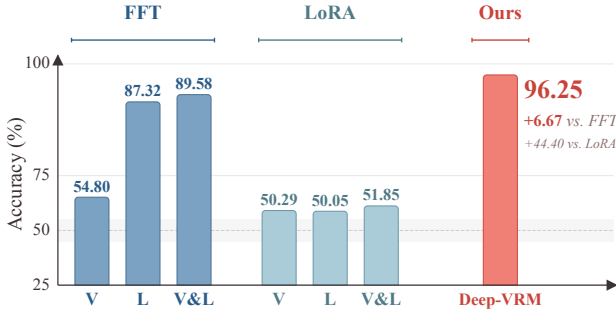


Figure 3. We finetuned pretrained Qwen-2.5-VL-7B (Bai et al., 2025) on a training set D_2 (for low-level artifacts). We applied Full Fine-Tuning (FFT) and LoRA-based adaptation to train ViT (V), Large Language Model (L), and ViT&Large Language Model (V&L). Our approach (with LoRA rank=64) achieves superior accuracy, demonstrating its capabilities in identifying generative artifacts. In contrast, standard LoRA tuning on the original MLLM architecture fails to capture these low-level artifacts effectively.

3.1. Analysis

To investigate the deficiencies of MLLMs in AIGI detection, we selected the pretrained Qwen-2.5-VL-7B as the backbone and analyzed its characteristics on the datasets: D_1 (focusing on semantics) and D_2 (focusing on low-level generator traces). Our analysis reveals that the pretrained MLLM, primarily optimizing vision-language alignment, naturally lacks sensitivity to the subtle low-level artifacts in D_2 . This creates a dilemma in representation learning: standard LoRA tuning on the original single-stream MLLM architecture is insufficient to recover these suppressed features, while full fine-tuning captures them but induces catastrophic forgetting of the original semantic knowledge (see Figure 3 and Table 2).

FINDING 1. Native MLLMs cannot learn generalizable low-level artifacts without aggressively modifying the feature space, which compromises the extraction of semantic features.

Therefore, balancing the learning of low-level artifacts with the preservation of semantic features is key to efficient MLLM-based detection. To identify the specific layers responsible for semantic processing, we examined the layer-wise contribution to detection using linear probes on Qwen-2.5-VL-7B. As shown in Figure 4, detection accuracy for

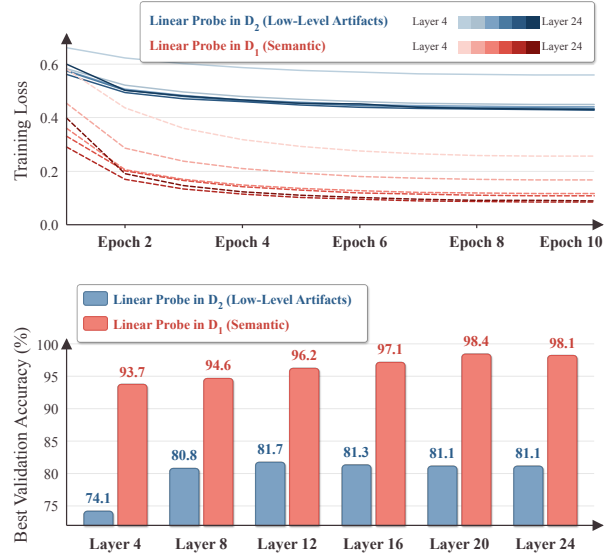


Figure 4. Layer-wise linear probing on Qwen-2.5-VL-7B. (Top) Training loss for D_1 (semantics-focused, dashed lines) converges faster and lower than for D_2 (artifact-focused, solid lines). (Bottom) Validation accuracy highlights a semantic convergence zone (layers 16–20) where D_1 performance peaks, while D_2 accuracy remains stagnant ($\approx 81\%$) across all depths.

semantic anomalies (D_1) improves rapidly and converges in the early-to-middle layers (1-16), whereas artifact detection (D_2) remains ineffective throughout. This indicates that these early-to-middle layers are the foundational “semantic convergence zone”.

FINDING 2. The early-to-middle layers of Native MLLMs are critical for forming semantic representations.

These findings explain why full fine-tuning compromises semantic capabilities: forcing the early layers to learn contradictory low-level artifact cues disrupts their established semantic convergence. Consequently, our strategy explicitly decouples this learning process. We preserve the early layers to maintain robust pre-formed semantic features and employ a Residual Injection strategy to introduce artifact features solely into the deep layers (post-convergence), where they can be integrated without interference.

3.2. Model Architecture

Our analysis (see Section 3.1) indicates a fundamental limitation: pre-trained MLLMs lack the inherent sensitivity to low-level artifacts necessary for AIGI detection. Although full fine-tuning can enable this capability, it comes at a high cost, resulting in the catastrophic forgetting of the model’s robust semantic representations. To address the problem, we propose **Deep Visual Residual MLLM (Deep-VRM)**, a streamlined architecture designed to heighten sensitivity to

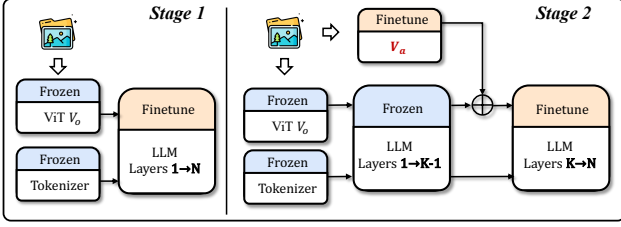


Figure 5. The architecture of our proposed method. The training process consists of two stages, where different parameters are finetuned in each stage. $\mathcal{V}_a$ is constructed by integrating an adapter into the base architecture of $\mathcal{V}_o$.

low-level artifacts while preserving robust semantic reasoning. Given an input image I and a text instruction sequence $\mathbf{x}_{1:N}$, the visual embeddings derived from the encoder $\mathcal{V}_o$ are concatenated with the textual embeddings to form the initial embedding sequence $\mathbf{H}^{(0)}$:

$$\mathbf{H}^{(0)} = [\mathbf{v}_{\text{emb}}, \mathbf{x}_1, \dots, \mathbf{x}_N], \quad \text{where } \mathbf{v}_{\text{emb}} = \mathcal{V}_o(I). \quad (1)$$

Subsequently, this input is processed by the frozen shallow layers of the LLM (denoted as LLM_{pre}). We define K as the first trainable LLM layer, i.e., the residual injection boundary. Thus, LLM_{pre} contains layers 1 to $K-1$ and yields the intermediate hidden states $\mathbf{H}^{(K-1)}$, which can be decomposed into visual and textual components:

$$\begin{aligned} \mathbf{H}^{(K-1)} &= \text{LLM}_{\text{pre}}(\mathbf{H}^{(0)}) \\ &= [\mathbf{h}_v^{(K-1)}, \mathbf{h}_{t,1}^{(K-1)}, \dots, \mathbf{h}_{t,N}^{(K-1)}]. \end{aligned} \quad (2)$$

Since these frozen early layers tend to suppress high-frequency artifact signals, we introduce a ‘‘Green Road’’ residual pathway for direct intervention. We equip the visual encoder $\mathcal{V}_o$ with lightweight LoRA adapters, denoting this adapted module as $\mathcal{V}_a$, to explicitly extract perceptual features from the image I and inject them into the visual embeddings $\mathbf{h}_v^{(K-1)}$ before layer K . This residual connection effectively bypasses the semantic extraction stage of the early-to-middle layers:

$$\tilde{\mathbf{h}}_v^{(K-1)} = \alpha \cdot \mathbf{h}_v^{(K-1)} + \beta \cdot \mathcal{V}_a(I), \quad (3)$$

where α and β are scaling factors (set to 0.5) that balance the semantic context with raw artifact cues. Finally, the enhanced visual tokens are re-concatenated with the original textual context and fed into the remaining layers (LLM_{post}) to generate the final response $\mathbf{O}$:

$$\begin{aligned} \tilde{\mathbf{H}}^{(K-1)} &= [\tilde{\mathbf{h}}_v^{(K-1)}, \mathbf{h}_{t,1}^{(K-1)}, \dots, \mathbf{h}_{t,N}^{(K-1)}], \\ \mathbf{O} &= \text{LLM}_{\text{post}}(\tilde{\mathbf{H}}^{(K-1)}). \end{aligned} \quad (4)$$

3.3. Training Strategy

We fine-tune the proposed MLLM using a standard autoregressive loss. Given an input image I and a target response

sequence $Y = \{y_1, y_2, \dots, y_L\}$, the model is optimized by minimizing the negative log-likelihood:

$$\mathcal{L}_{SFT} = - \sum_{i=1}^L \log P(y_i | I, y_{<i}; \Theta)$$

where Θ denotes the set of trainable parameters. As illustrated in Figure 5, our approach consists of two stages:

Stage 1: Semantic Alignment and Prior Activation. In this stage, we employ D_1 to fine-tune the LLM component, while keeping all other modules frozen. This stage is designed to activate and align the LLM’s inherent pre-trained knowledge, enabling it to perform AIGI detection based on high-level semantic knowledge.

Stage 2: Artifact-Aware Refinement. We integrate both D_1 and D_2 to train the residual injection architecture. To preserve the discriminative priors established in the first stage, the first $K-1$ LLM layers remain frozen. The residual features extracted by $\mathcal{V}_a$ are injected before layer K , after which $\mathcal{V}_a$ and the LLM layers from K to N are optimized. This design enables artifact-aware refinement while preserving the semantic representations formed by the frozen early layers.

4. Experiments

Implementation We employ Qwen-2.5-VL-7B (Bai et al., 2025) as the default backbone. All images are resized to match an equivalent pixel budget of 512×512 while preserving their original aspect ratios during training. We train the model for 2 epochs using AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.95$, and weight decay $1e^{-3}$, following a cosine learning-rate decay schedule. The learning rates are set to $1e^{-6}$ for the projector and $1e^{-4}$ for the visual encoder and LLM parameters. We apply LoRA (Hu et al., 2022) with rank $r = 64$ and LoRA alpha 128. The adapted visual branch $\mathcal{V}_a$ is initialized from the frozen visual encoder $\mathcal{V}_o$ and equipped with LoRA adapters; therefore, compared with the standard baseline MLLM, our method introduces no additional trainable parameters beyond the LoRA parameters. For the default Qwen-2.5-VL-7B setting, the total number of trainable parameters is 115.02M.

Evaluation Metric and Comparison Methods In evaluation, following previous works (Yan et al., 2025d; Chen et al., 2025a), we use balanced accuracy for benchmarks containing real and fake images. For fake-only benchmarks, we report the fake-class accuracy. To comprehensively verify the effectiveness of our method, we also provide the results of the following models: Xception (Chollet, 2017), CNNSpot (Wang et al., 2020), F3Net (Qian et al., 2020), GramNet (Liu et al., 2020), UnivFD (Ojha

Table 3. The generalizable performance (ACC (%)) in GenImage Dataset (Zhu et al., 2023).

Model	MidJ	SDv1.4	SDv1.5	ADM	GLIDE	Wukong	VQDM	BigGAN	AVG
Xception	57.97	98.06	97.98	51.16	57.51	97.79	50.34	48.74	69.94
CNNSpot	61.25	98.13	97.54	51.50	55.13	93.51	51.83	51.06	69.99
F3Net	52.26	99.30	99.21	49.64	50.46	98.70	45.56	49.59	68.09
GramNet	63.00	94.19	94.22	48.69	46.19	93.79	49.20	44.71	66.75
UnivFD	77.29	97.01	96.67	50.94	78.47	91.52	65.72	55.91	76.69
NPR	62.00	99.75	99.64	56.79	82.69	97.89	54.43	52.26	75.68
AIDE	79.38	99.74	99.76	78.54	91.82	98.65	80.26	66.89	86.88
DIRE	51.11	55.07	55.31	49.93	50.02	53.71	49.87	49.85	51.86
DRCT/Conv-B	94.43	99.37	99.19	66.42	73.31	99.25	76.85	59.41	83.53
DRCT/UnifD	85.82	92.33	91.87	75.18	87.44	92.23	89.12	87.38	87.67
OMAT	90.36	97.52	97.46	83.82	97.41	97.62	95.53	97.34	94.63
Deep-VRM	96.16	99.65	99.42	89.82	99.08	99.44	97.29	98.50	97.42

Table 4. Generalizable performance (ACC (%)) on evaluation set in SynthBuster (Bammey, 2024). The methods marked with * indicate results reported in the original paper. This dataset includes fake images only.

Method	Glide	SD1.3	SD1.4	SD2	SD XL	MidJ	DALLE2	DALLE3	Firefly	AVG
UnivFD*	10.10	24.30	21.80	34.40	21.50	0.00	42.40	0.00	61.70	14.30
C2P-CLIP	12.00	51.10	54.20	39.10	56.20	6.50	12.00	27.20	19.70	30.89
DecCLIP	48.30	93.30	93.40	68.16	54.10	30.00	89.44	0.40	71.50	60.96
DRCT	14.10	89.60	88.20	99.90	89.60	99.40	4.10	35.60	11.40	59.10
PatchShuffle	80.40	100.00	100.00	82.49	77.70	82.80	19.80	8.10	13.50	62.75
Deep-VRM	99.90	99.80	100.00	71.90	95.80	96.80	99.90	99.90	86.50	94.50

et al., 2023), NPR (Tan et al., 2024b), AIDE (Yan et al., 2025c), DIRE (Wang et al., 2023), DRCT (Chen et al., 2024), OMAT (Zhou et al., 2025a), AIGI-Holmes (Zhou et al., 2025b), SAFE (Li et al., 2025a), C2P-CLIP (Tan et al., 2025), FatFormer (Liu et al., 2024), CO-SPY (Cheng et al., 2025), PatchShuffle (Zheng et al., 2024), Forensic-Chat (Lin et al., 2025b).

4.1. Evaluation for Generalization

Performance on the AIGI detection benchmarks To verify the generalization of our method, we conduct experiments to evaluate the performance on the following datasets: GenImage (Zhu et al., 2023), GenImage++ (Zhou et al., 2025a), AIGI-Holmes (Zhou et al., 2025b), EvalGen (Chen et al., 2025a), CommunityAI (Park & Owens, 2025), SynthBuster (Bammey, 2024), AIGI-NOW (Chen et al., 2026). As detailed in Tables 3, 4, 7 and Table 17, 18, 19, 20 in Appendix, our method consistently achieves SOTA performance. This consistent superiority across diverse generative sources underscores the effectiveness of Deep-VRM.

4.2. Evaluation for Performance in Wild Scenario

Considering practical applications, validation on ‘in-the-wild’ datasets is necessary. Real-world images often suffer from complex post-processing (e.g., re-compression, re-sizing) and originate from diverse, unknown sources. To address this, we selected WildRF (Cavia et al., 2024) and AIGI-Bench (Li et al., 2025c), which comprise data collected from social media platforms (e.g., Facebook, Reddit, Twitter) and online communities. As shown in Table 5, our method consistently outperforms SOTA competitors by a significant margin. This validates our model’s robustness

Table 5. Performance (ACC (%)) on WildRF (Cavia et al., 2024) and AIGI-Bench (Li et al., 2025c), two benchmarks designed to reflect wild scenario and evaluate the robustness of detectors.

Method	WildRF				AIGI Bench		
	FaceBook	Reddit	Twitter	AVG	SocialRF	CommunityAI	AVG
FatFormer	64.38	76.65	40.00	60.34	57.98	50.62	54.30
CO-SPY	50.00	56.79	73.30	60.03	55.54	53.02	54.28
C2P-CLIP	54.38	68.40	47.27	56.68	53.13	50.98	52.06
SAFE	62.50	61.70	40.33	54.84	58.00	54.25	56.13
AIDE	75.00	55.48	48.00	59.49	57.80	54.15	55.98
Deep-VRM	85.94	91.07	91.53	89.51	83.68	97.28	90.48

 Table 6. Ablation Study to explore the impact to vary the insert depth K of visual residual features.

Depth	GenImage	SynthBuster	AIGI-Bench	WildRF	AVG
4	97.05	96.69	83.68	74.82	88.06
8	96.83	94.99	86.65	74.68	88.29
12	97.26	93.40	91.06	89.12	92.91
16	97.39	94.50	90.25	89.51	92.97
20	96.42	94.44	90.30	89.20	92.59

and its ability to generalize effectively to unconstrained scenarios where low-level artifacts are often degraded.

4.3. Comparison for other MLLMs

To verify the effectiveness of our proposed method, we conducted a comprehensive comparison against representative MLLMs. The baselines include general-purpose models (e.g., Qwen-2-VL (Wang et al., 2024), Qwen-2.5-VL (Bai et al., 2025), LLaVA-1.5 (Liu et al., 2023)) and a specialized forensic MLLM (Forensic-Chat) tailored for AI-generated image detection. In Table 8, it can be observed that general-purpose MLLMs struggle with the detection task, yielding an average accuracy ranging from 50% to 68%. While the domain-specific Forensic-Chat (Lin et al., 2025b) achieves a respectable average accuracy of 88.75%, our method achieves the best performance with an average accuracy of **93.28%**. Notably, on the challenging benchmarks AIGI-Bench and WildRF, our method surpasses the strongest baseline by margins of 8.29% and 5.57%, respectively. This demonstrates that injecting visual residual features effectively bridges the gap between general understanding and low-level artifact detection.

4.4. Ablation Study

Impact of Visual Residual Depth. We analyze the impact of feature injection depth on model performance (Table 6). The results corroborate the representation conflict hypothesis discussed in Sec. 1. Shallow injection (Depth 4) forces the model to prioritize low-level artifacts prematurely, which boosts accuracy on specific uncompressed datasets (e.g., SynthBuster) but disrupts the early encoding of semantic features, leading to poor generalization on compressed data (e.g., WildRF). Conversely, deep injection (Depth 16)

Table 7. The generalizable performance (ACC (%)) in AIGI-Now Dataset (Chen et al., 2026).

Method	Nano Banana		GPT-4o		Jimeng		Kling		Minimax		Flux Pro		Flux Krea		Flux Dev		Flux Kontext		AVG	
	Pix	Sem	Pix	Sem	Pix	Sem	Pix	Sem	Pix	Sem	Pix	Sem	Pix	Sem	Pix	Sem	Pix	Sem	Pix	Sem
NPR	84.4	49.9	91.2	50.1	41.9	49.9	88.3	49.9	52.1	50.0	42.7	50.0	52.3	50.0	89.3	50.0	79.2	49.9	69.0	50.0
UnivFD	51.0	51.4	53.8	53.3	50.7	52.3	51.4	53.9	50.5	52.1	51.9	52.0	51.0	52.2	63.6	49.9	57.5	55.9	52.0	52.6
FatFormer	53.4	49.9	49.9	49.8	48.4	49.9	54.8	50.0	49.1	50.0	48.4	50.0	49.5	50.0	52.4	50.0	63.6	49.9	52.1	49.9
SAFE	99.3	50.1	99.7	50.4	50.0	50.5	99.6	50.0	49.8	50.1	49.9	50.5	51.2	50.1	99.8	75.3	98.8	83.4	77.5	56.7
C2P-CLIP	50.0	50.0	49.8	50.0	49.8	50.0	50.7	49.9	50.3	50.0	52.0	50.0	50.1	50.0	52.6	50.0	60.7	50.0	51.7	50.0
AIDE	98.9	51.8	74.7	53.5	63.9	51.4	98.2	55.4	51.4	54.1	60.1	53.8	50.4	56.9	99.1	59.0	97.9	80.6	77.1	57.4
DRCT	68.9	57.7	75.0	57.0	89.3	57.7	66.9	56.8	78.4	57.1	83.0	58.2	91.6	58.2	86.5	58.6	86.9	55.5	80.7	57.4
AlignedForensics	88.0	50.3	50.8	50.8	82.1	50.0	69.6	50.2	56.7	50.3	65.5	49.9	59.5	50.1	78.6	50.2	65.0	50.0	68.4	50.2
CO-SPY	74.1	58.3	78.9	63.1	83.0	76.2	88.3	78.8	77.9	65.6	77.3	72.7	89.1	74.0	88.5	74.9	65.1	66.0	80.2	70.0
BFree	69.6	52.8	59.2	56.7	75.6	54.5	86.1	60.7	56.2	49.9	71.8	54.5	59.5	51.5	74.8	54.5	77.1	53.4	70.0	54.3
Deep-VRM	99.9	99.0	100.0	97.0	99.4	96.5	99.9	97.3	93.6	92.4	96.2	97.3	94.4	88.5	99.1	96.3	96.7	79.60	97.7	93.7

Table 8. Comparison of performance (ACC%) across different MLLMs on AIGI detection benchmarks. Forensic-Chat is an MLLM-based detector tailored for AIGI detection.

Method	GenImage	AIGI-Now	AIGI Bench	WildRF	AVG
Qwen-2-VL-2B	56.54	56.66	57.97	62.42	58.40
Qwen-2-VL-7B	56.66	67.56	66.43	83.94	68.65
Qwen-2.5-VL-3B	63.93	62.38	69.48	78.69	68.62
Qwen-2.5-VL-7B	50.18	57.54	53.78	54.64	54.04
LLaVA-1.5-7B	49.84	50.17	50.89	51.41	50.58
Forensic-Chat	97.55	94.21	82.19	81.08	88.75
Deep-VRM	97.42	95.70	90.48	89.51	93.28

Table 9. Ablation Study to explore the impact of different stages.

Model	GenImage	WildRF	AIGI-Bench	SynthBuster	AVG
Stage 1	95.34	73.27	69.55	79.86	79.51
Stage 1 + Stage 2	97.42	89.51	90.48	94.50	92.98

preserves the integrity of the semantic latent space in the early layers while introducing forensic signals at a stage where the model can perform joint reasoning. This configuration yields a significant robustness gain (+14% on WildRF compared to Depth 4) while maintaining competitive precision on pixel-level artifacts, motivating our choice of Depth 16 as the default setting.

Ablation on Different Training Stages We conducted experiments to investigate the individual contributions of different training stages to the overall performance. Table 9 compares the results of the initial training phase (Stage 1) against the full training pipeline (Stage 1 + Stage 2). Compared with the first stage alone, the integration of Stage 2 yields a significant performance gain.

Ablation on Freezing Early LLM Layers For our new architecture, we investigate the effect of freezing early LLM layers versus fine-tuning all layers. The results in Table 10 demonstrate that while fine-tuning all layers (No) performs comparably on standard datasets, our strategy of freezing early layers (Yes) significantly boosts performance on challenging in-the-wild benchmarks such as WildRF and AIGI-Bench. Consequently, our method achieves a higher average accuracy of 92.98%, validating that freezing early layers ef-

Table 10. Ablation Study on Freezing the Early LLM Layers.

Freezing	GenImage	WildRF	AIGI-Bench	SynthBuster	AVG
No	97.95	83.49	87.39	98.62	91.86
Yes (Ours)	97.42	89.51	90.48	94.50	92.98

Table 11. Ablation Study on altering different backbones in our method.

Model	GenImage	WildRF	AIGI-Bench	SynthBuster	AVG
Qwen-2-VL-7B	97.28	85.41	89.11	96.08	91.97
LLaVA-1.5-7B	92.02	92.47	92.48	96.48	93.36
Qwen-2.5-VL-3B	85.72	71.52	80.80	92.54	82.65
Qwen-2.5-VL-7B	97.42	89.51	90.48	94.50	92.98

fectively prevents the forgetting of general semantic knowledge required for robust detection.

Ablation on Altering Backbone To evaluate the generalization capability of our proposed method, we apply it across various MLLM architectures, including Qwen-2.5-VL-7B, Qwen-2-VL-7B (Wang et al., 2024), and LLaVA-1.5-7B (Liu et al., 2023). As shown in Table 11, our approach consistently delivers robust performance across multiple benchmarks, demonstrating its architecture-agnostic effectiveness. However, we observe a performance degradation in the Qwen-2.5-VL-3B model. We attribute this primarily to its reduced parameter scale, which limits its capacity to capture complex semantic features. Furthermore, our use of a fixed LoRA rank ($r = 64$) may have induced underfitting, as the trainable parameter budget decreases proportionally with the model’s hidden dimension.

Ablation on Residual Fusion Coefficients To analyze the impact of the residual fusion coefficients, we vary α and β in Eq. 3. In our implementation, α scales the pre-formed intermediate visual-token representation $\mathbf{h}_v^{(K-1)}$, while β scales the injected artifact-specific features $\mathcal{V}_a(I)$. As shown in Table 12, the balanced setting $(\alpha, \beta) = (0.5, 0.5)$ achieves the best average performance. When α is very small, the pre-formed representation is strongly down-weighted, leading to clear drops, especially on GenImage and SynthBuster. This

Table 12. Ablation study on the residual fusion coefficients α and β in Eq. 3.

(α, β)	GenImage	SynthBuster	AIGI-Bench	WildRF	AVG
(0.1, 0.9)	94.94	84.42	90.67	89.04	89.77
(0.3, 0.7)	94.29	90.02	91.48	90.31	91.53
(0.5, 0.5)	97.39	94.50	90.25	89.51	92.91
(0.7, 0.3)	96.76	94.24	90.10	86.97	92.76
(0.9, 0.1)	96.78	92.82	91.21	90.04	92.71

 Table 13. Performance comparison of expert detection methods, the baseline MLLM, and ours when trained on the combined D_1 and D_2 datasets.

Method	GenImage	WildRF	AIGI-Bench	SynthBuster	AVG
UnivFD	78.10	70.15	82.15	53.15	70.89
AIDE	93.19	78.93	82.14	87.02	85.32
Qwen-2.5-VL-7B	92.07	70.85	55.22	69.99	72.03
Deep-VRM	97.42	89.51	90.48	94.50	92.98

suggests that preserving a sufficient intermediate representation is important for stable fusion. However, once α reaches 0.5, further increasing its value brings limited additional gains, and the average performance becomes nearly saturated. This indicates that the model is relatively robust to moderate changes in the fusion coefficients, likely because the trainable modules can partially adapt to different feature scales during optimization. These observations suggest that the fusion coefficients mainly affect the relative scale of the two branches, rather than imposing a structural constraint as the injection depth does. Thus, this ablation should be viewed as a sensitivity analysis of the fusion design, showing that $(\alpha, \beta) = (0.5, 0.5)$ offers a simple and effective trade-off between preserving the pre-formed semantic representation and introducing artifact-aware residual cues.

Impact for Dataset To decouple the influence of training data, we retrained representative expert detection methods (UnivFD and AIDE) and the baseline MLLM (Qwen-2.5-VL-7B) using the combined datasets D_1 and D_2 . For the baseline MLLM, we employ LoRA fine-tuning with a rank of 64. These models were subsequently evaluated on diverse real-world benchmarks to assess their generalization capabilities. As presented in Table 13, even when provided with the same high-quality forensic data, Deep-VRM significantly outperforms existing approaches across all metrics, particularly on “in-the-wild” datasets like WildRF. This confirms that the performance gains are primarily attributed to our deep residual injection architecture rather than the dataset scale alone.

Comparison with Different Feature Fusion Paradigms. To isolate the source of Deep-VRM’s improvement, we compare it with several variants that use the same adapted visual branch $\mathcal{V}_a$ but integrate it into the MLLM in different ways, as shown in Table 14. This comparison is intended to exam-

Table 14. Validation of different feature fusion paradigms. Feature-Concat uses a linear classifier, Input Fusion follows the external expert paradigm by directly inserting auxiliary visual tokens into the input sequence, and Early/Late Fusion correspond to residual injection at shallow/deep layers.

Method	GenImage	SynthBuster	AIGI-Bench	WildRF	AVG
Feature-Concat	50.59	5.38	51.58	48.71	39.07
Input Fusion	96.93	95.71	84.81	79.88	89.33
Early Fusion ($K=4$)	97.05	96.69	83.68	74.82	88.06
Late Fusion ($K=20$)	96.42	94.44	90.30	89.20	92.59
Deep-VRM ($K=16$)	97.39	94.50	90.25	89.51	92.91

ine whether our gains come from the proposed deep residual injection design, rather than from merely introducing extra visual features. First, we replace our residual injection with a Feature-Concat baseline, where the features from $\mathcal{V}_a$ are pooled and concatenated with the final LLM hidden state for classification:

$$\hat{y} = \text{MLP}([\mathbf{h}_L; \text{Pool}(\mathcal{V}_a(I))]). \quad (5)$$

Although this baseline has access to the same adapted visual branch, its performance drops substantially, indicating that simply attaching artifact-aware features to a final classifier cannot effectively couple low-level forensic cues with the MLLM representation. We then compare Deep-VRM with Input Fusion, which follows the common external expert paradigm by treating $\mathcal{V}_a$ as an auxiliary visual expert and directly inserting its unpooled visual tokens into the input sequence:

$$\mathbf{H}_{\text{IF}}^{(0)} = [\mathcal{V}_o(I), \mathcal{V}_a(I), \mathbf{x}_1, \dots, \mathbf{x}_N], \quad (6)$$

where $\mathcal{V}_a(I)$ has the same token format as $\mathcal{V}_o(I)$ and is inserted without pooling. Compared with Deep-VRM, this design exposes all subsequent LLM layers to additional expert tokens and therefore substantially increases the input sequence length, computation, and memory cost. However, despite this heavier token-level fusion, Input Fusion remains clearly inferior to Deep-VRM, especially on the in-the-wild benchmarks WildRF and AIGI-Bench. This shows that the advantage of Deep-VRM is not simply due to providing more expert-like visual tokens to the MLLM; rather, the location and manner of feature integration are critical. Finally, we evaluate whether the residual pathway should be injected earlier or later than our default setting. Early Fusion ($K=4$) injects artifact features before stable semantic representations are formed and performs poorly on WildRF, suggesting that premature intervention can disturb semantic processing. Late Fusion ($K=20$) avoids such early interference but leaves fewer subsequent layers for joint reasoning over semantic and artifact-aware signals. By injecting $\mathcal{V}_a(I)$ at $K=16$, Deep-VRM achieves the best average accuracy, supporting our design choice of introducing artifact-aware residual features after semantic convergence while still preserving sufficient depth for multimodal reasoning.

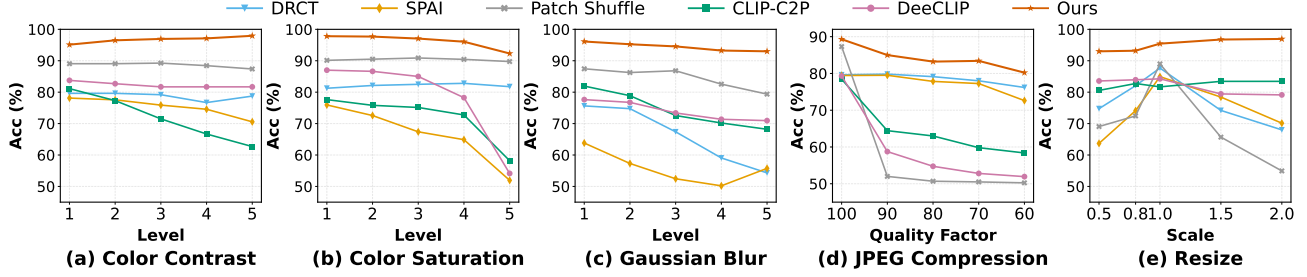


Figure 6. Robust performance (Acc (%)) on GenImage after post-process. We deploy 5 different post-process to verify the robustness of our method and compared methods.

4.5. Evaluation for Robustness

In this experiment, we evaluate the robustness of Deep-VRM on GenImage under common image degradations, with results shown in Figure 6. Since real-world images are often post-processed during uploading, sharing, and storage, such perturbations can weaken fragile pixel-level traces and challenge practical AIGI detection. We consider five representative degradations, including Color Contrast, Color Saturation, Gaussian Blur, JPEG compression, and Resize, covering both photometric changes and spatial/frequency distortions. Compared with existing models, Deep-VRM maintains consistently higher accuracy across degradation types and severity levels. This suggests that Deep-VRM does not rely on a single fragile artifact pattern, but jointly exploits low-level cues and semantic detection features through artifact-aware residual injection. Detailed degradation configurations are provided in the Appendix.

4.6. Visualization

Although Deep-VRM is not designed to enhance explainability, we visualize conversational examples of Deep-VRM in Figure 7. The results demonstrate that our model does not merely provide binary classification but effectively articulates multi-level generative traces. In these open-ended responses, the model grounds its judgment in observable visual evidence rather than relying only on the final “real” or “fake” token. Specifically, as shown in the zebra example, the model identifies low-level artifacts rooted in abnormal digital textures. Conversely, in the vehicle example, it detects semantic-level anomalies that violate real-world logic, such as the distorted license plate and the missing side mirror. This confirms that our strategy enables the MLLM to autonomously prioritize crucial features—whether subtle textures or complex semantic inconsistencies—to provide evidence-based detection.

5. Conclusion

In this paper, we presented Deep Visual Residual MLLM (Deep-VRM), a novel architecture designed to achieve full-



Figure 7. Visualization of conversational AIGI detection. Deep-VRM pinpoint low-level artifacts (e.g., digital textures in the zebra image) and semantic-level flaws (e.g., the missing mirror and distorted plate in the car image).

spectrum forensic signal perception by resolving the representation conflict between semantic reasoning and low-level artifact detection in MLLMs. Our layer-wise analysis reveals a critical functional stratification: while early-to-middle layers of pretrained MLLMs are essential for semantic convergence, forcing them to learn generator-specific traces leads to catastrophic forgetting of semantic knowledge. To address this, Deep-VRM employs a Deep Residual Injection strategy that preserves the frozen semantic processing of early layers while injecting artifact-specific visual features directly into the intermediate layers of the LLM. Extensive evaluations across diverse benchmarks—including GenImage, AIGI-Now, and challenging “in-the-wild” datasets like WildRF—demonstrate that our method achieves state-of-the-art performance without relying on external expert detectors. Furthermore, Deep-VRM exhibits superior robustness to real-world image degradations such as compression and resizing. By effectively bridging the gap between high-level semantic analysis and low-level forensic perception, Deep-VRM provides a unified and generalizable framework for reliable AI-generated image detection.

Acknowledgement

Dr Bin Li was supported in part by NSFC and in part by Shenzhen R&D Program (Grant JCYJ20250604181211016, SYSPG20241211174032004).

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J., et al. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- Bammey, Q. Synthbuster: Towards detection of diffusion model generated images. *IEEE Open Journal of Signal Processing*, 5:1–9, 2024. doi: 10.1109/OJSP.2023.3337714.
- Cavia, B., Horwitz, E., Reiss, T., and Hoshen, Y. Real-time deepfake detection in the real-world. *arXiv preprint arXiv:2406.09398*, 2024.
- Chen, B., Zeng, J., Yang, J., and Yang, R. Drct: Diffusion reconstruction contrastive training towards universal detection of diffusion generated images. In *Forty-first International Conference on Machine Learning*, 2024.
- Chen, R., Xi, J., Yan, Z., Zhang, K.-Y., Wu, S., Xie, J., Chen, X., Xu, L., Guan, I., Yao, T., and Ding, S. Dual data alignment makes ai-generated image detector easier generalizable. In *Advances in Neural Information Processing Systems*, volume 38, pp. 106475–106500, 2025a.
- Chen, R., Gao, J., Lin, K., Zhang, K., Zhao, Y., Guan, I., Yao, T., and Ding, S. Aligngemini: Generalizable ai-generated image detection through task-model alignment, 2026.
- Chen, Y., Yan, Z., Cheng, G., Zhao, K., Lyu, S., and Wu, B. X2-dfd: A framework for explainable and extendable deepfake detection. In *Advances in Neural Information Processing Systems*, volume 38, pp. 83792–83839, 2025b.
- Cheng, S., Lyu, L., Wang, Z., Zhang, X., and Sehwal, V. Co-spy: Combining semantic and pixel features to detect synthetic images by ai. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 13455–13465, 2025.
- Chollet, F. Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1251–1258, 2017.
- Chu, B., Xu, X., Wang, X., Zhang, Y., You, W., and Zhou, L. Fire: Robust detection of diffusion-generated images via frequency-guided reconstruction error. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 12830–12839, 2025.
- Duan, H., Yang, J., Qiao, Y., Fang, X., Chen, L., Liu, Y., Dong, X., Zang, Y., Zhang, P., Wang, J., et al. Vlmevalkit: An open-source toolkit for evaluating large multi-modality models. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pp. 11198–11201, 2024.
- Gao, Y., Chang, D., Yu, B., Qin, H., Chen, L., Liang, K., and Ma, Z. Fakereasoning: Towards generalizable forgery detection and reasoning. *arXiv preprint arXiv:2503.21210*, 2025.
- Guillaro, F., De Rosa, V., Cozzolino, D., and Verdoliva, L. Quality-aware calibration for ai-generated image detection in the wild. *arXiv preprint arXiv:2604.15027*, 2026.
- Hartman, M., Jayaraman, V., Choraria, M., Bhimaraju, A., and Varshney, L. R. Skip-it? theoretical conditions for layer skipping in vision-language models. *arXiv preprint arXiv:2509.25584*, 2025.
- He, X., Zhou, Y., Fan, B., Li, B., Zhu, G., and Ding, F. V1-forgery face triad: Detection, localization and attribution via multimodal large language models. In *Advances in Neural Information Processing Systems*, volume 38, pp. 163010–163044, 2025.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.
- Jia, S., Lyu, R., Zhao, K., Chen, Y., Yan, Z., Ju, Y., Hu, C., Li, X., Wu, B., and Lyu, S. Can chatgpt detect deepfakes? a study of using multimodal large language models for media forensics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4324–4333, 2024.
- Jiang, Z., Chen, J., Zhu, B., Luo, T., Shen, Y., and Yang, X. Devils in middle layers of large vision-language models: Interpreting, detecting and mitigating object hallucinations via attention lens. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 25004–25014, 2025.
- Kang, H., Wen, S., Wen, Z., Ye, J., Li, W., Feng, P., Zhou, B., Wang, B., Lin, D., Zhang, L., et al. Legion: Learning to ground and explain for synthetic image detection. In

- Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 18937–18947, 2025.
- Karageorgiou, D., Papadopoulos, S., Kompatsiaris, I., and Gavves, E. Any-resolution ai-generated image detection by spectral learning. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 18706–18717, 2025.
- Li, B., Xiao, X., Li, W., Lin, K., Zhang, X., Tian, B., and Zhao, Y. Texture-adaptive cost modeling via residual frequency quantization and predictive order awareness for webp steganography. *IEEE Transactions on Circuits and Systems for Video Technology*, 2026.
- Li, O., Cai, J., Hao, Y., Jiang, X., Hu, Y., and Feng, F. Improving synthetic image detection towards generalization: An image transformation perspective. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*, pp. 2405–2414, 2025a.
- Li, S., Guo, Y., Chen, S., Li, B., Lin, K., Chen, C., Li, H., Yao, T., and Ding, S. Ditl2: Dual-stage invariance transfer learning for generalizable document image tampering localization. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 82–91, 2025b.
- Li, Z., Yan, J., He, Z., Zeng, K., Jiang, W., Xiong, L., and Fu, Z. Is artificial intelligence generated image detection a solved problem? In *Advances in Neural Information Processing Systems*, volume 38, 2025c.
- Lin, K., Lin, Y., Li, W., Yao, T., and Li, B. Standing on the shoulders of giants: Reprogramming visual-language model for general deepfake detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 5262–5270, 2025a.
- Lin, K., Yan, Z., Chen, R., Ye, J., Zhang, K.-Y., Zhou, Y., Jin, P., Li, B., Yao, T., and Ding, S. Seeing before reasoning: A unified framework for generalizable and explainable fake image detection. *arXiv preprint arXiv:2509.25502*, 2025b.
- Lin, K., Yan, Z., Zhang, K.-Y., Hao, L., Zhou, Y., Lin, Y., Li, W., Yao, T., Ding, S., and Li, B. Guard me if you know me: Protecting specific face-identity from deep-fakes. In *Advances in Neural Information Processing Systems*, volume 38, pp. 138174–138212, 2025c.
- Lin, T.-Y. et al. Microsoft coco: Common objects in context. In *European Conference on Computer Vision*, pp. 740–755. Springer, 2014.
- Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- Liu, H., Tan, Z., Tan, C., Wei, Y., Wang, J., and Zhao, Y. Forgery-aware adaptive transformer for generalizable synthetic image detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10770–10780, 2024.
- Liu, R., Han, Y., Zhang, Z., Yao, L., Yan, Z., Shen, J., Chen, Z., Sun, B., Weng, L., Dong, J., et al. Beyond artifacts: Real-centric envelope modeling for reliable ai-generated image detection. *arXiv preprint arXiv:2512.20937*, 2025.
- Liu, R., Cui, M., Qin, Z., Yan, Z., Chen, R., Han, Y., Li, Z., Chen, J., Chen, Z., Lin, K., et al. Mirror: Manifold ideal reference reconstructor for generalizable ai-generated image detection. *arXiv preprint arXiv:2602.02222*, 2026.
- Liu, Z. et al. Global texture enhancement for fake face detection in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 8060–8069, 2020.
- Nguyen, T., Khan, N., Tran, K., Phan, N., and Khalil, I. Prpo: Paragraph-level policy optimization for vision-language deepfake detection. *arXiv preprint arXiv:2509.26272*, 2025.
- Ojha, U. et al. Towards universal fake image detectors that generalize across generative models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 24480–24489, 2023.
- Park, J. and Owens, A. Community forensics: Using thousands of generators to train fake image detectors. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 8245–8257, 2025.
- Peng, S., Wang, Z., Gao, L., Zhu, X., Zhang, T., Liu, A., Zhang, H., and Lei, Z. Mllm-enhanced face forgery detection: A vision-language fusion solution. *arXiv preprint arXiv:2505.02013*, 2025.
- Qian, Y. et al. Thinking in frequency: Face forgery detection by mining frequency-aware clues. In *European Conference on Computer Vision*, pp. 86–103. Springer, 2020.
- Russakovsky, O. et al. Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015.
- Shen, J., Zheng, J., Xue, Y., Chen, H., Yao, Y., Kang, H., Liu, R., Gong, H., Yang, Y., Wang, D., et al. Dino-detect: A simple yet effective framework for blur-robust ai-generated image detection. *arXiv preprint arXiv:2511.12511*, 2025.

- Skean, O., Arefin, M. R., Zhao, D., Patel, N. N., Naghiyev, J., LeCun, Y., and Shwartz-Ziv, R. Layer by layer: Uncovering hidden representations in language models. In *Forty-second International Conference on Machine Learning*, 2025.
- Tan, C., Zhao, Y., Wei, S., Gu, G., Liu, P., and Wei, Y. Frequency-aware deepfake detection: Improving generalizability through frequency space domain learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 5052–5060, 2024a.
- Tan, C., Zhao, Y., Wei, S., Gu, G., Liu, P., and Wei, Y. Rethinking the up-sampling operations in cnn-based generative network for generalizable deepfake detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 28130–28139, 2024b.
- Tan, C., Tao, R., Liu, H., Gu, G., Wu, B., Zhao, Y., and Wei, Y. C2p-clip: Injecting category common prompt in clip to enhance generalization in deepfake detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 7184–7192, 2025.
- Unsplash. Access the world’s largest open library dataset. <https://unsplash.com/data>, 2025. Accessed on: Jan. 29, 2026.
- Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.
- Wang, S.-Y., Wang, O., Zhang, R., Owens, A., and Efros, A. A. Cnn-generated images are surprisingly easy to spot... for now. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8695–8704, 2020.
- Wang, Z., Bao, J., Zhou, W., Wang, W., Hu, H., Chen, H., and Li, H. Dire for diffusion-generated image detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 22445–22455, 2023.
- Wen, S., ye, j., Feng, P., Kang, H., Wen, Z., Chen, Y., Wu, J., wu, w., He, C., and Li, W. Spot the fake: Large multimodal model-based synthetic image detection with artifact explanation. In *Advances in Neural Information Processing Systems*, volume 38, pp. 58972–59005, 2025.
- Xu, Z., Zhang, X., Huang, Q., Zhou, X., and Zhang, J. Avatarshield: Visual reinforcement learning for human-centric synthetic video detection. *arXiv preprint arXiv:2505.15173*, 2025.
- Xu, Z., Zhang, X., Xu, Y., Huang, Q., Chen, S., Yao, T., Ding, S., and Zhang, J. Genshield: Unified detection and artifact correction for ai-generated images. *arXiv preprint arXiv:2605.16122*, 2026.
- Yan, J., Li, Z., Wang, F., Wang, B., He, Z., and Fu, Z. Dgs-net: Distillation-guided gradient surgery for clip fine-tuning in ai-generated image detection. *arXiv preprint arXiv:2511.13108*, 2025a.
- Yan, J., Wang, F., Jiang, W., Li, Z., and Fu, Z. Ns-net: Decoupling clip semantic information through null-space for generalizable ai-generated image detection. *arXiv preprint arXiv:2508.01248*, 2025b.
- Yan, J., Li, Z., Wang, F., He, Z., and Fu, Z. Dual frequency branch framework with reconstructed sliding windows attention for ai-generated image detection. *IEEE Transactions on Information Forensics and Security*, 2026.
- Yan, S., Li, O., Cai, J., Hao, Y., Jiang, X., Hu, Y., and Xie, W. A sanity check for ai-generated image detection. In Yue, Y., Garg, A., Peng, N., Sha, F., and Yu, R. (eds.), *International Conference on Learning Representations*, volume 2025, pp. 70702–70720, 2025c.
- Yan, Z., Wang, J., Jin, P., Zhang, K.-Y., Liu, C., Chen, S., Yao, T., Ding, S., Wu, B., and Yuan, L. Orthogonal subspace decomposition for generalizable AI-generated image detection. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267, pp. 70268–70288. PMLR, 2025d.
- Ye, J., Jiang, D., Wang, Z., Zhu, L., Hu, Z., Huang, Z., He, J., Yan, Z., Yu, J., Li, H., et al. Echo-4o: Harnessing the power of gpt-4o synthetic images for improved image generation. *arXiv preprint arXiv:2508.09987*, 2025.
- Yu, Y., Zhou, Y., Li, B., Lin, K., Li, H., Ni, J., and Cao, B. Agentfox: Llm agent-guided fusion with explainability for ai-generated image detection. *arXiv preprint arXiv:2603.23115*, 2026.
- Zhao, Y., Huang, J., Hu, J., Wang, X., Mao, Y., Zhang, D., Jiang, Z., Wu, Z., Ai, B., Wang, A., et al. Swift: a scalable lightweight infrastructure for fine-tuning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 29733–29735, 2025.
- Zheng, C., Lin, C., Zhao, Z., Wang, H., Guo, X., Liu, S., and Shen, C. Breaking semantic artifacts for generalized ai-generated image detection. *Advances in Neural Information Processing Systems*, 37:59570–59596, 2024.
- Zhou, Y., He, X., Lin, K., Fan, B., Ding, F., and Li, B. Breaking latent prior bias in detectors for generalizable aigc image detection. In *Advances in Neural Information Processing Systems*, volume 38, pp. 30649–30679, 2025a.
- Zhou, Y., He, X., Lin, K., Fan, B., Ding, F., and Li, B. Simplicity prevails: The emergence of generalizable aigi detection in visual foundation models. *arXiv preprint arXiv:2602.01738*, 2026.

Zhou, Z., Luo, Y., Wu, Y., Sun, K., Ji, J., Yan, K., Ding, S., Sun, X., Wu, Y., and Ji, R. Aigi-holmes: Towards explainable and generalizable ai-generated image detection via multimodal large language models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 18746–18758, 2025b.

Zhu, C., Wang, M., Liu, J., Chang, C.-C., and Echizen, I. Evoguard: An extensible agentic rl-based framework for practical and evolving ai-generated image detection. *arXiv preprint arXiv:2603.17343*, 2026.

Zhu, M., Chen, H., Yan, Q., Huang, X., Lin, G., Li, W., Tu, Z., Hu, H., Hu, J., and Wang, Y. Genimage: A million-scale benchmark for detecting ai-generated image. *Advances in Neural Information Processing Systems*, 36: 77771–77782, 2023.

A. Experiments

Evaluation Setting of Deep-VRM In this paper, we conducted all performance evaluations using binary-choice prompts. Specifically, we standardized the input textual prompt as: Is this image real or fake? Please just answer "real" or "fake". The model’s accuracy was then calculated by parsing the generated output for the target tokens “real” or “fake”.

Comparison with Other PEFT Methods To further examine whether the improvement comes merely from the choice of parameter-efficient fine-tuning (PEFT) method, we compare the standard LoRA baseline with OLoRA under the same rank setting ($r = 64$). All variants are trained on D_2 and evaluated on GenImage. As shown in Table 15, OLoRA improves over the standard LoRA baseline, but it still largely trails Deep-VRM. This indicates that mitigating parameter-space interference at the PEFT level alone is insufficient for learning low-level forensic artifacts in MLLMs. In contrast, Deep-VRM explicitly changes the feature integration pathway by injecting artifact-aware residual visual features after semantic convergence, leading to substantially stronger generalization.

Table 15. Ablation study of different PEFT methods. All methods are trained on D_2 with rank $r = 64$ and evaluated on GenImage.

Method	GenImage
LoRA	51.85
OLoRA	60.40
Ours (LoRA)	96.25

Performance on More AIGI detection benchmarks We supplement more evaluation of multiple benchmarks in Table 17, 18, and 20.

Detail Setting in Robust Evaluation In Table 16, we exhibit the setting in Robust Evaluation. The degradation includes Color Contrast adjustment via $P_{out} = \alpha P_{in}$ and Color Saturation modulation via $C_{out} = 0.5 + \beta(C_{in} - 0.5)$, where α and β represent the contrast scaling factor and chroma gain coefficient, respectively.

Table 16. Degradation configurations for robustness evaluation. Gaussian blurring was defined by kernel size (K) and standard deviation (σ); Color contrast was adjusted by the linear scaling factor (α); Color saturation was modulated by the chroma gain coefficient (β) in YCbCr space.

Level	Color Contrast	Gaussian Blur	Color Saturation
1	0.850	$K = (7, 7), \sigma = 1$	0.4
2	0.725	$K = (9, 9), \sigma = 2$	0.3
3	0.600	$K = (13, 13), \sigma = 3$	0.2
4	0.475	$K = (17, 17), \sigma = 4$	0.1
5	0.350	$K = (21, 21), \sigma = 5$	0.0

Table 17. The generalizable performance (ACC (%)) in GenImage++ Dataset (Zhou et al., 2025a). This dataset includes fake images only.

Model	Flux	Flux-M	Flux-P	Flux-R	SD1.5-M	SDXL-M	SD3	SD3-P	SD3-R	AVG
Xception	36.86	10.48	4.65	5.45	97.27	20.63	38.00	5.83	15.06	26.03
CNNSpot	37.38	6.89	8.71	5.28	84.41	34.79	47.70	7.48	25.55	28.69
F3Net	25.18	7.79	2.83	7.90	94.15	24.01	46.67	0.84	30.28	26.63
GramNet	37.83	16.71	8.01	19.71	96.49	28.65	48.55	8.33	55.71	35.55
NPR	35.38	13.19	8.48	19.41	93.63	15.40	32.38	12.45	27.58	28.66
SPSL	67.13	16.55	43.76	25.73	71.14	17.74	44.58	16.22	29.75	36.96
SRM	8.46	2.92	0.37	1.93	96.62	6.39	9.97	0.55	4.43	14.63
DRCT/Conv-B	73.02	51.91	54.72	66.40	100.00	77.19	79.10	82.93	76.58	73.54
DRCT/UniFD	71.08	63.97	46.83	62.42	99.19	64.84	72.28	70.70	73.55	69.43
OMAT	96.53	92.55	97.60	97.67	100.00	99.17	98.27	90.38	98.82	96.78
Deep-VRM	99.85	99.95	100.00	100.00	97.33	98.98	99.20	100.00	99.55	99.43

Table 18. Generalizable performance (ACC (%)) on EvalGen (Chen et al., 2025a). We obtained the experimental results from the original paper. This dataset includes fake images only.

Method	Flux	GoT	Infinity	OmiGen	NOVA	AVG
UnivFD	4.00	9.20	15.70	8.30	39.60	15.40
FatFormer	9.90	47.80	44.70	98.30	27.30	45.60
C2P-CLIP	8.70	49.40	35.10	86.40	14.80	38.90
AIDE	16.20	21.60	4.00	14.90	18.40	15.00
AlignedForensics	45.00	84.40	79.60	90.80	85.20	77.00
DDA	87.00	99.30	99.50	99.50	100.00	94.00
Deep-VRM	100.00	100.00	100.00	100.00	100.00	100.00

Table 19. Performance (ACC (%)) on evaluation set in Community Forensics (Park & Owens, 2025). We ignored the subset ‘DALLE2’ in this table.

Method	DALLE3	DFGAN	Flux-dev	GALIP	Hourglass	IdeogramV1	IdeogramV2	Imagen3
C2P-CLIP	64.15	99.30	60.25	74.44	68.70	51.45	50.80	50.38
DeeCLIP	91.00	99.85	57.70	86.35	65.75	67.55	62.15	78.43
DRCT	94.80	50.65	88.70	53.50	52.40	92.05	90.80	93.38
PatchShuffle	99.85	66.15	98.25	51.70	65.65	96.90	96.20	98.85
Deep-VRM	99.80	99.95	99.90	81.75	96.55	99.80	99.35	99.91

Method	Kandinsky	Kvikontent	LCM-SD15	LCM-SDXL	LCM-SSD1B	MidJourney V5	Stable Cascade	AVG
C2P-CLIP	58.00	85.05	82.30	54.05	89.10	54.14	65.60	67.18
DeeCLIP	80.15	97.85	55.95	55.65	56.85	75.97	94.60	75.05
DRCT	99.75	98.65	95.85	97.90	89.85	97.82	97.90	86.27
PatchShuffle	99.50	99.30	92.55	71.30	58.70	81.51	94.40	84.72
Deep-VRM	99.95	99.70	99.55	96.30	96.10	99.97	99.90	97.90

Table 20. Generalization performance (ACC %) on the AIGI-Holmes dataset (Zhou et al., 2025b). Baseline results are cited from the original paper. Note that AIGI-Holmes* denotes the standalone MLLM setting, excluding the ensemble with external dedicated detectors.

Model	Janus	J-Pro-1B	J-Pro-7B	Show-o	LlamaGen	Infinity	VAR	PixArt-XL	SD3.5 L	FLUX	AVG
CNNSpot	70.00	70.90	85.00	72.20	61.90	86.80	59.90	78.20	63.80	79.90	72.90
AntiFakePrompt	72.20	84.30	84.80	86.20	96.20	83.60	90.70	81.70	92.80	66.10	83.90
UnivFD	87.60	96.90	96.40	85.90	93.10	79.20	64.30	75.70	87.80	69.60	83.60
NPR	51.20	69.50	73.90	93.70	93.50	93.80	85.90	93.40	91.60	93.60	84.00
LaRE	70.80	74.70	95.60	80.00	91.60	77.90	98.80	82.20	94.10	84.30	85.00
RINE	89.90	98.70	97.20	98.80	99.10	99.20	85.00	98.90	97.80	97.10	96.20
AIDE	91.20	98.90	97.80	98.00	99.40	98.70	93.60	98.60	99.40	94.40	97.00
AIGI-Holmes*	80.20	91.90	89.60	98.00	98.00	98.40	76.00	98.50	97.80	94.20	92.30
Deep-VRM	77.40	96.12	96.98	99.95	99.98	99.95	91.45	99.93	99.62	99.67	96.11